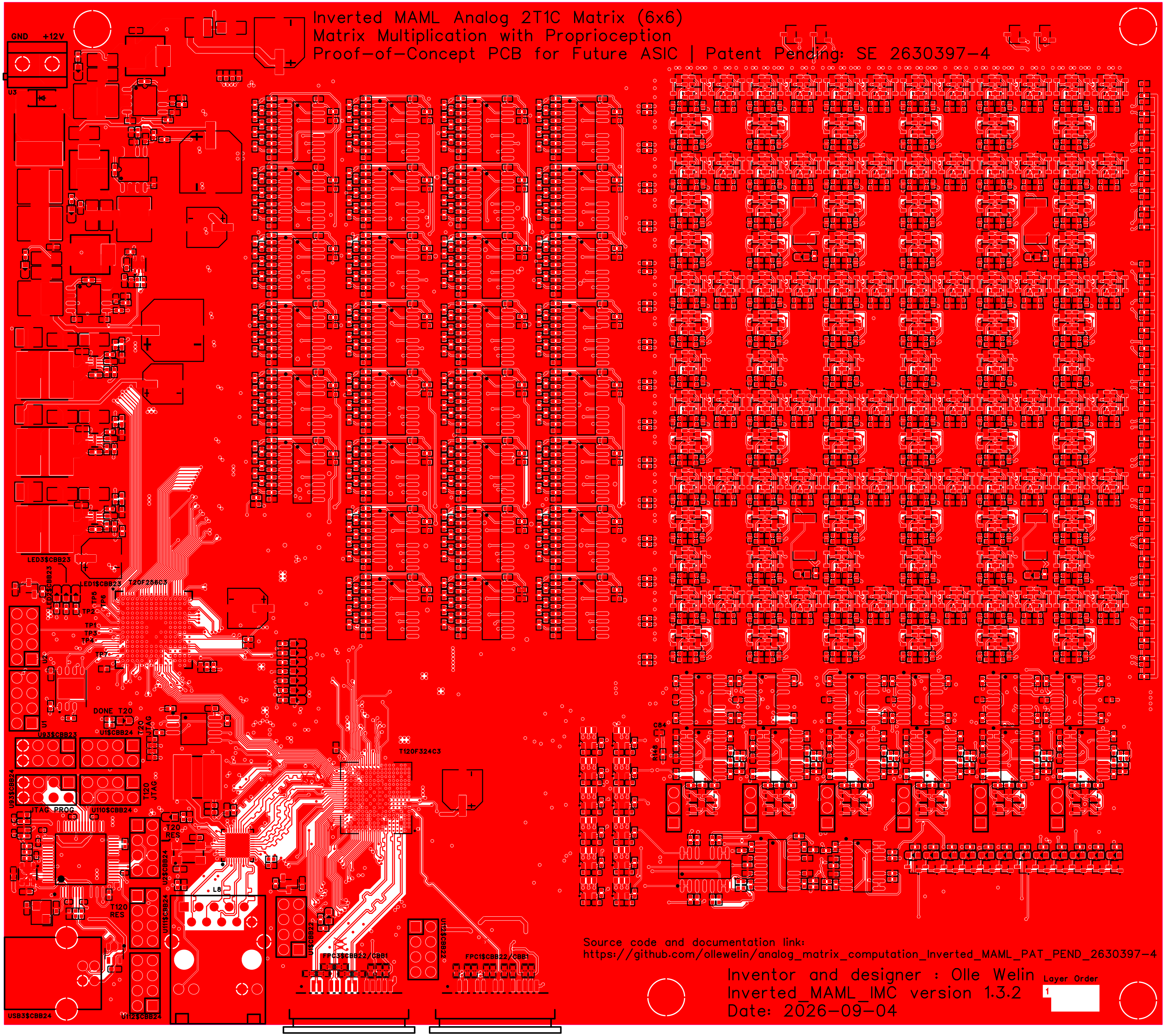
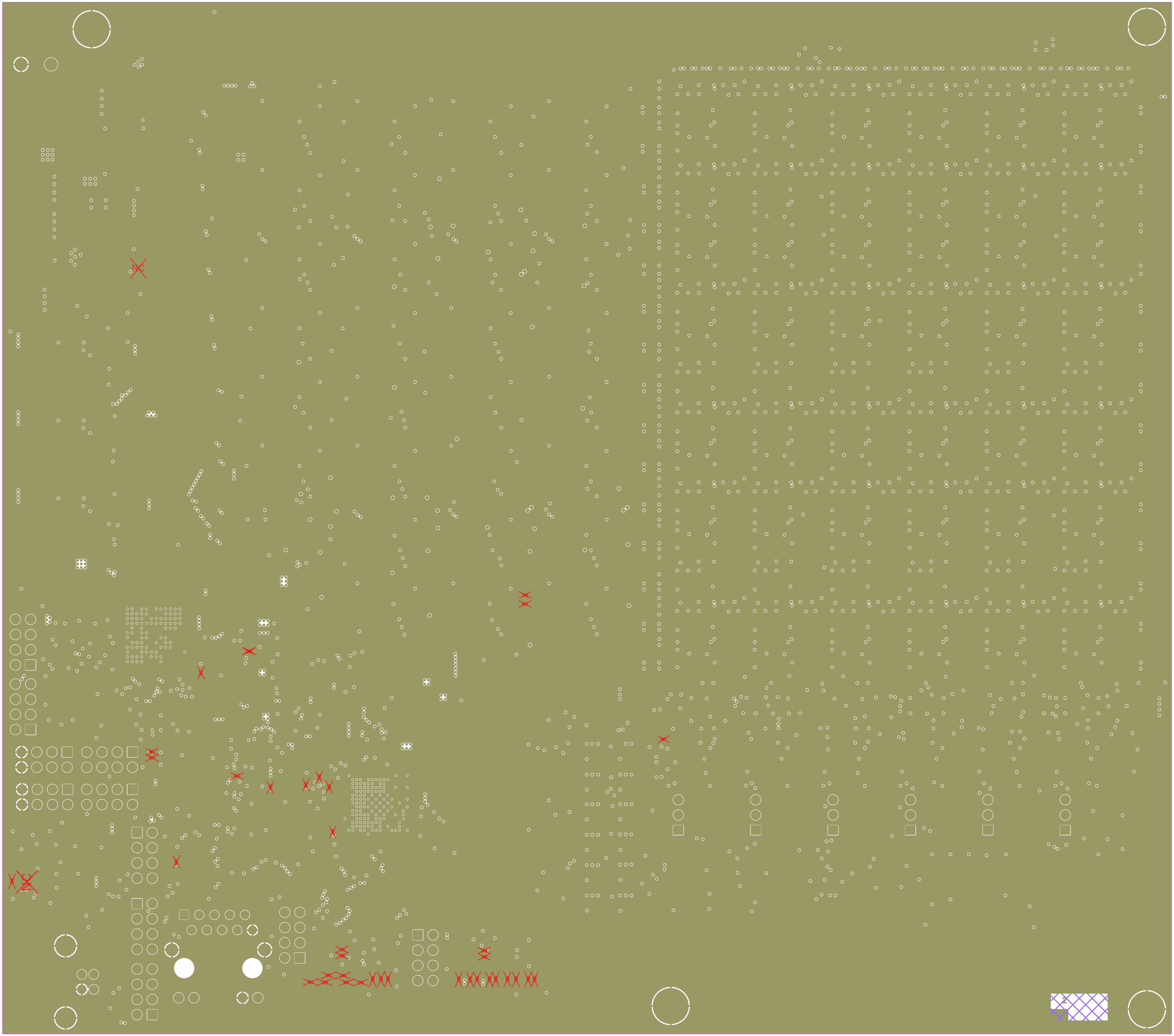

Inverted MAML Analog 2T1C Matrix (6x6)
Matrix Multiplication with Proprioception
Proof-of-Concept PCB for Future ASIC | Patent Pending: SE 2630397-4
GND +12V
JTAG PROG
Source code and documentation link:
https://github.com/ollewelin/analog_matrix_computation_Inverted_MAML_PAT_PEND_2630397-4
Inventor and designer : Olle Welin
Inverted_MAML_IMC version 1.3.2
Date: 2026-09-04
Layer Order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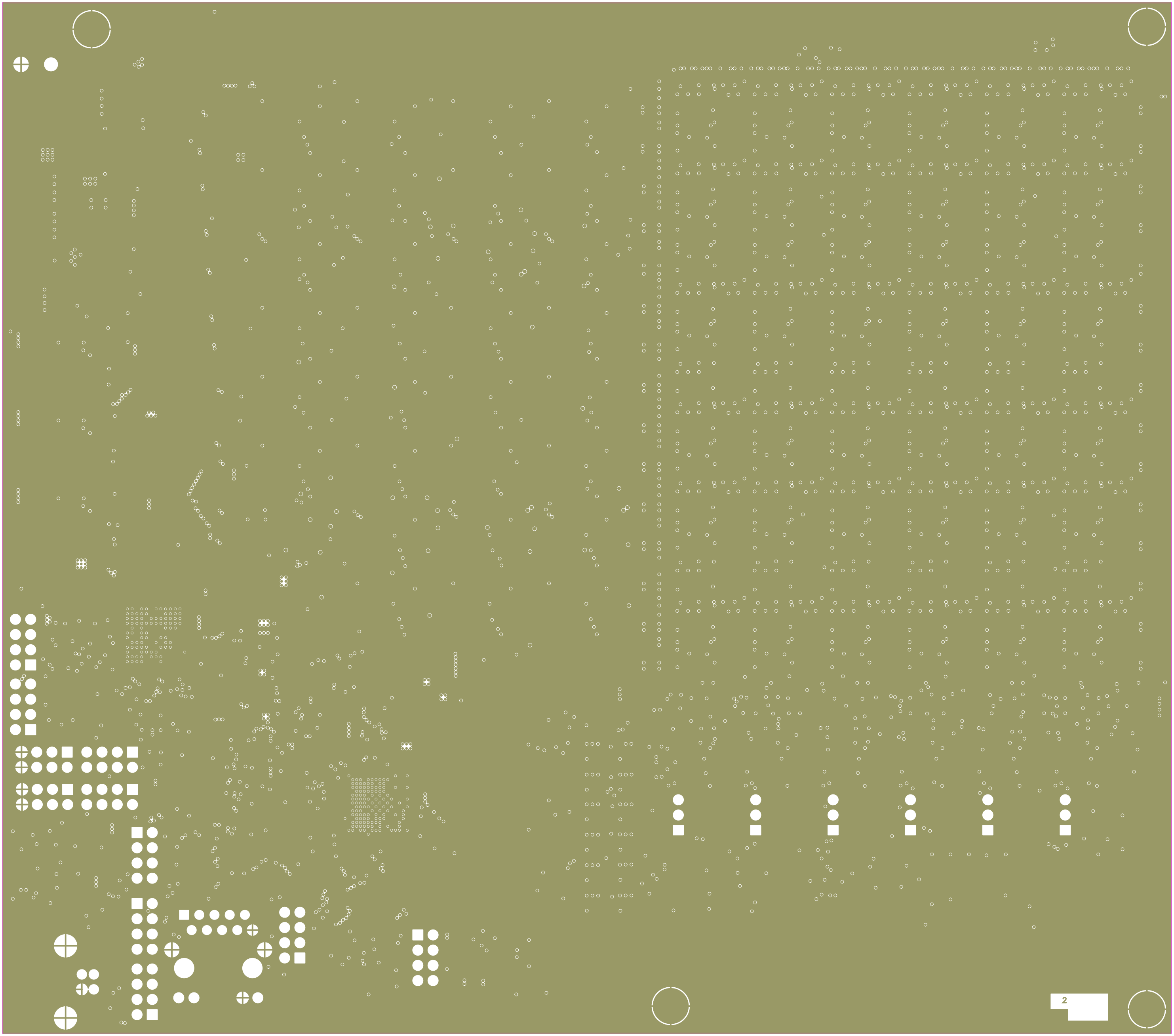
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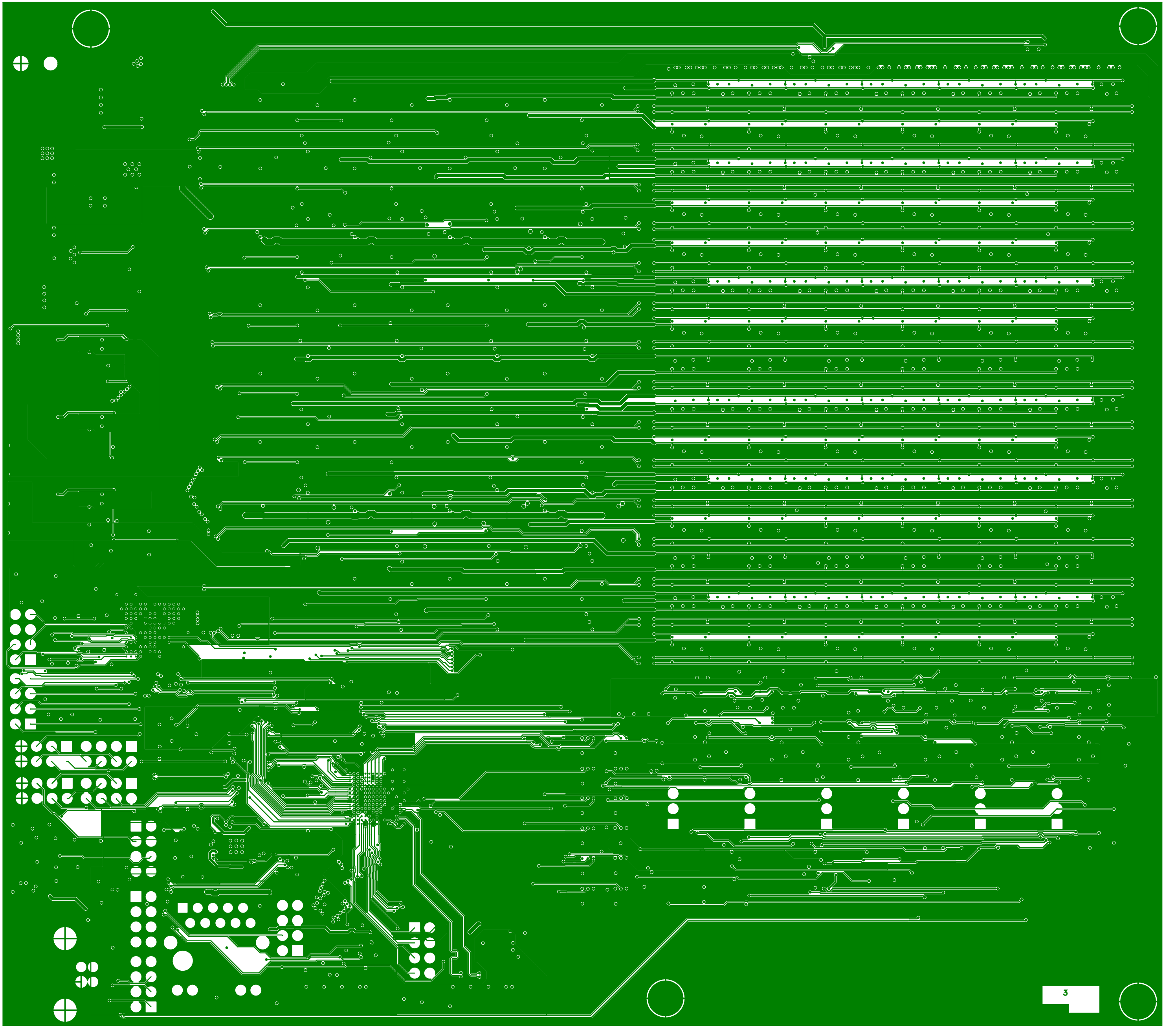
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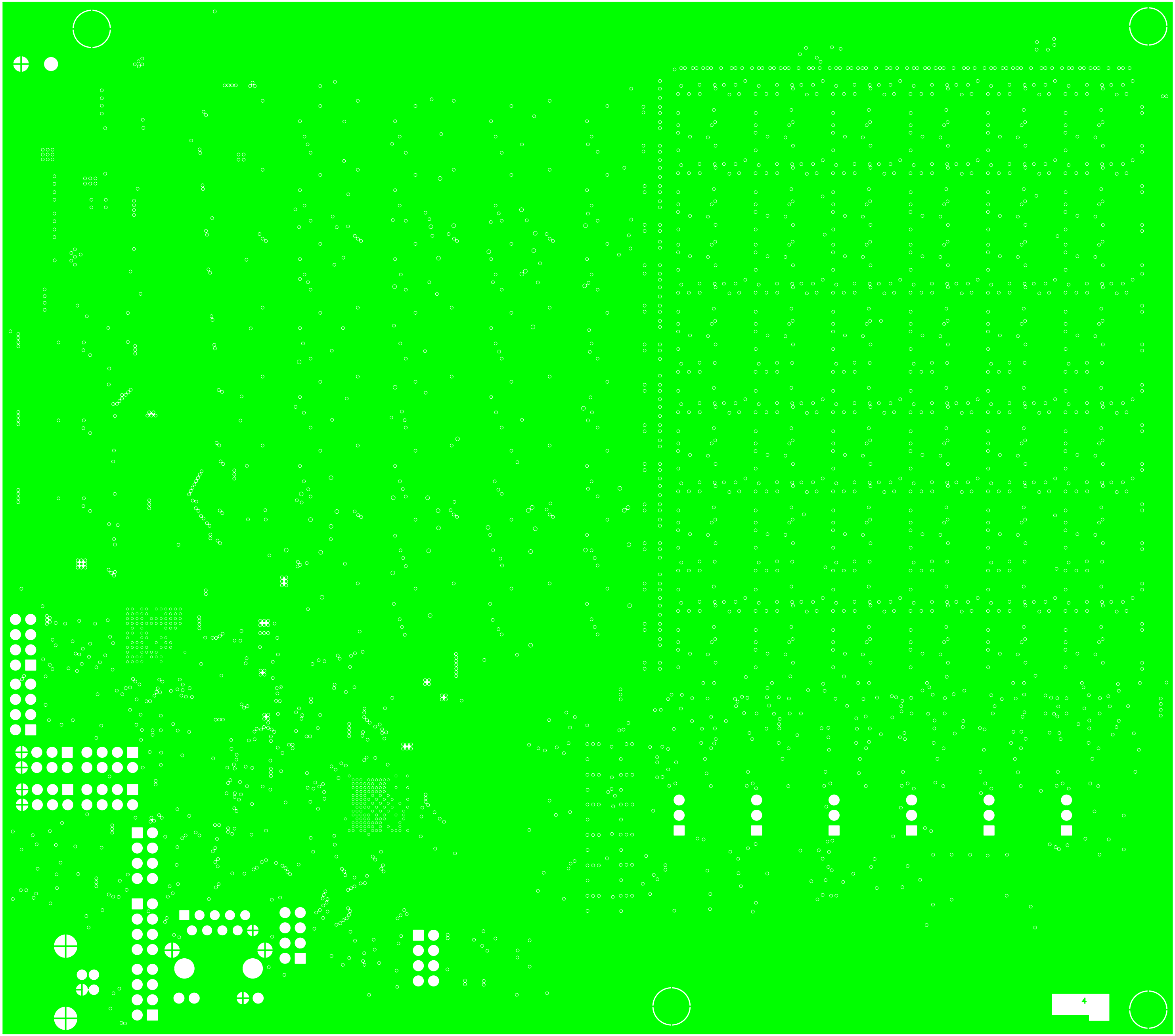
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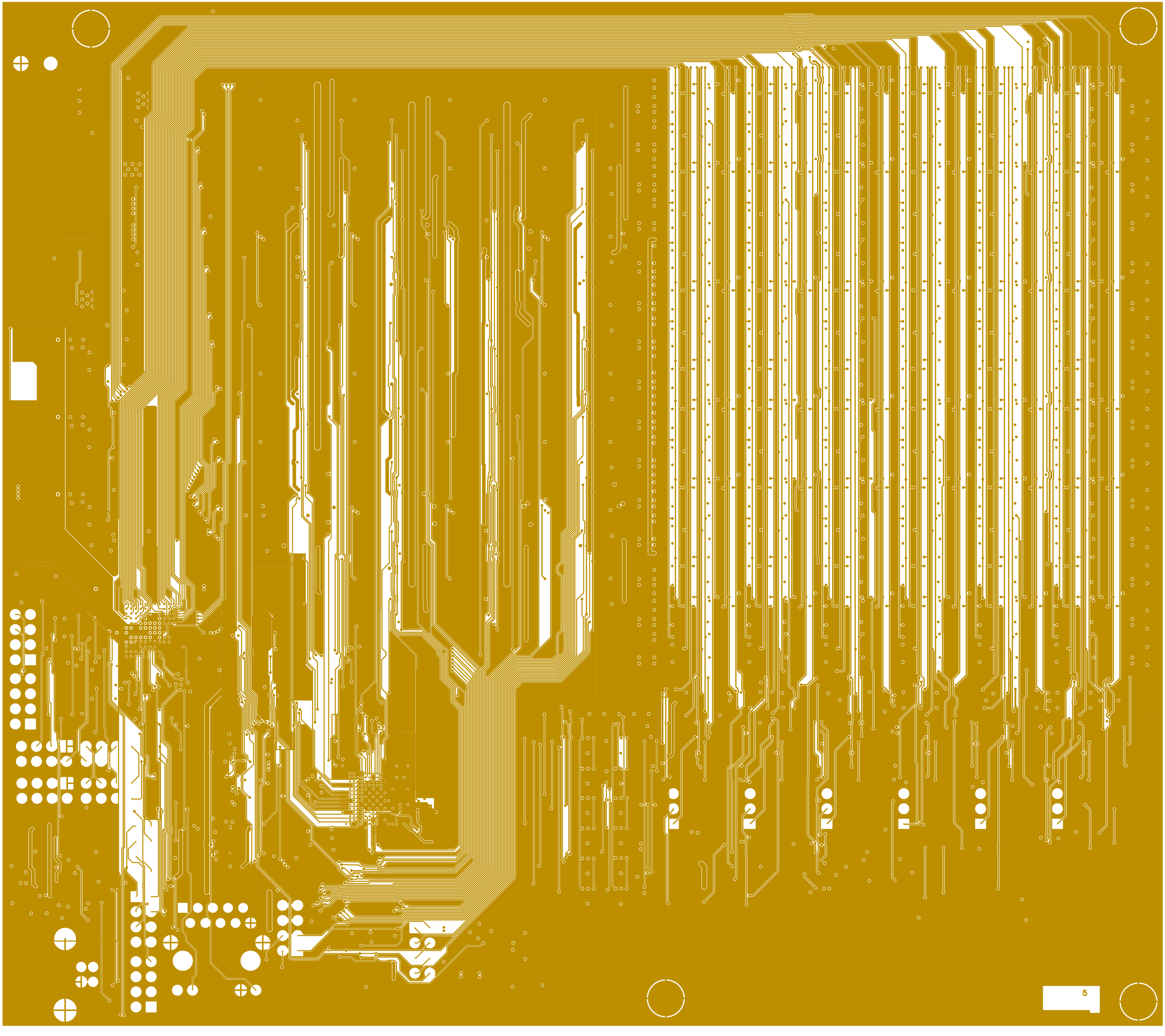
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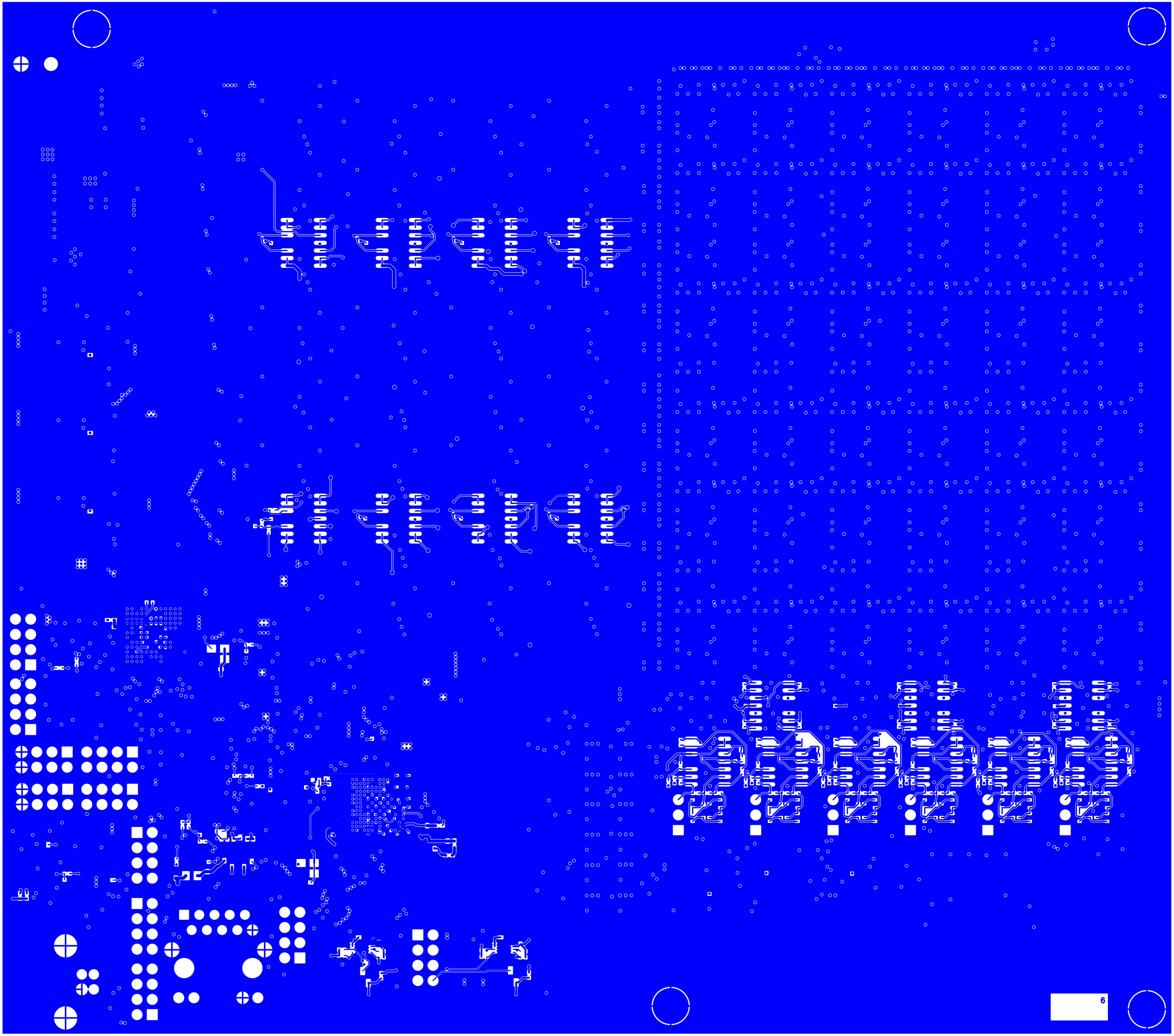
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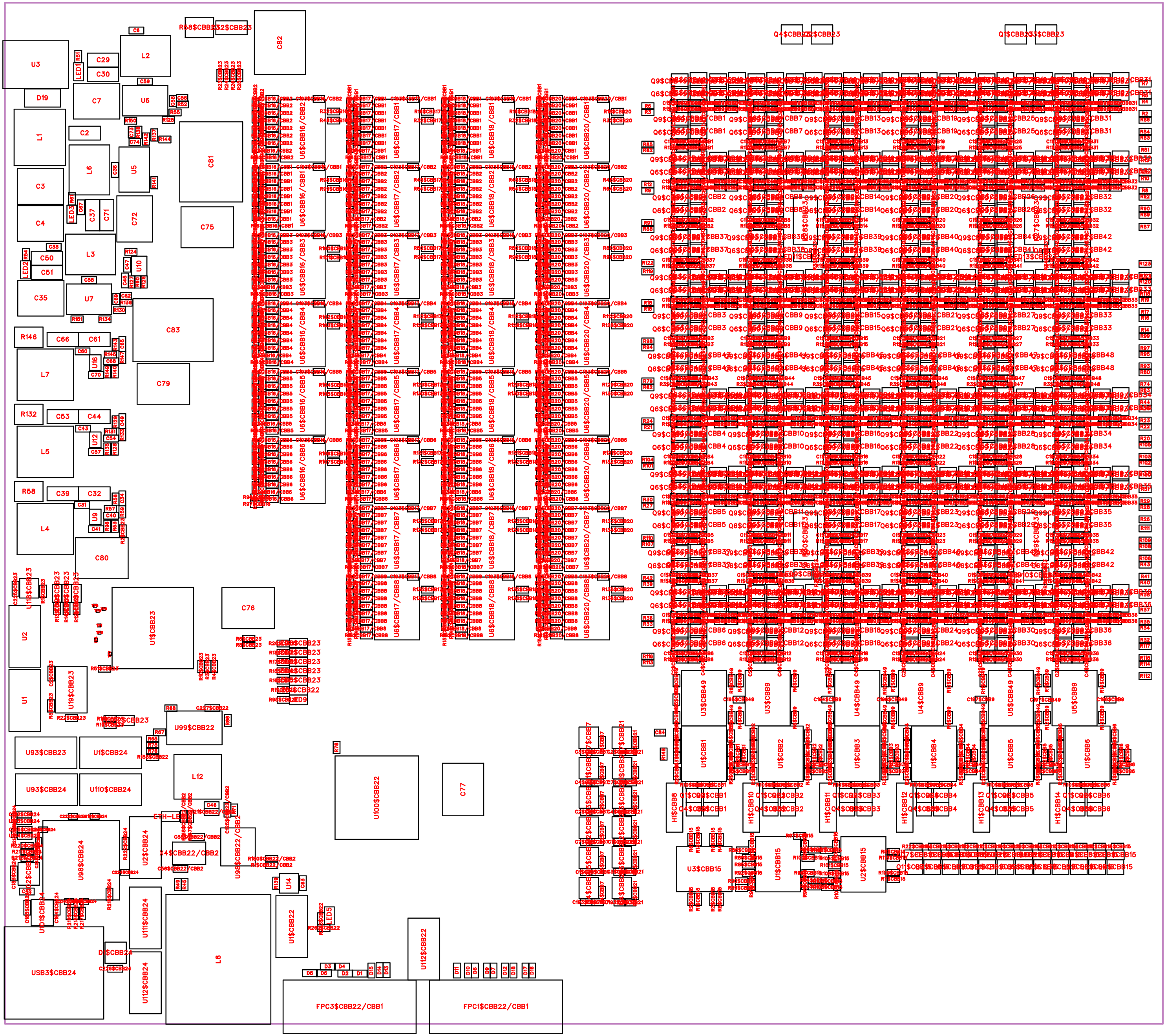

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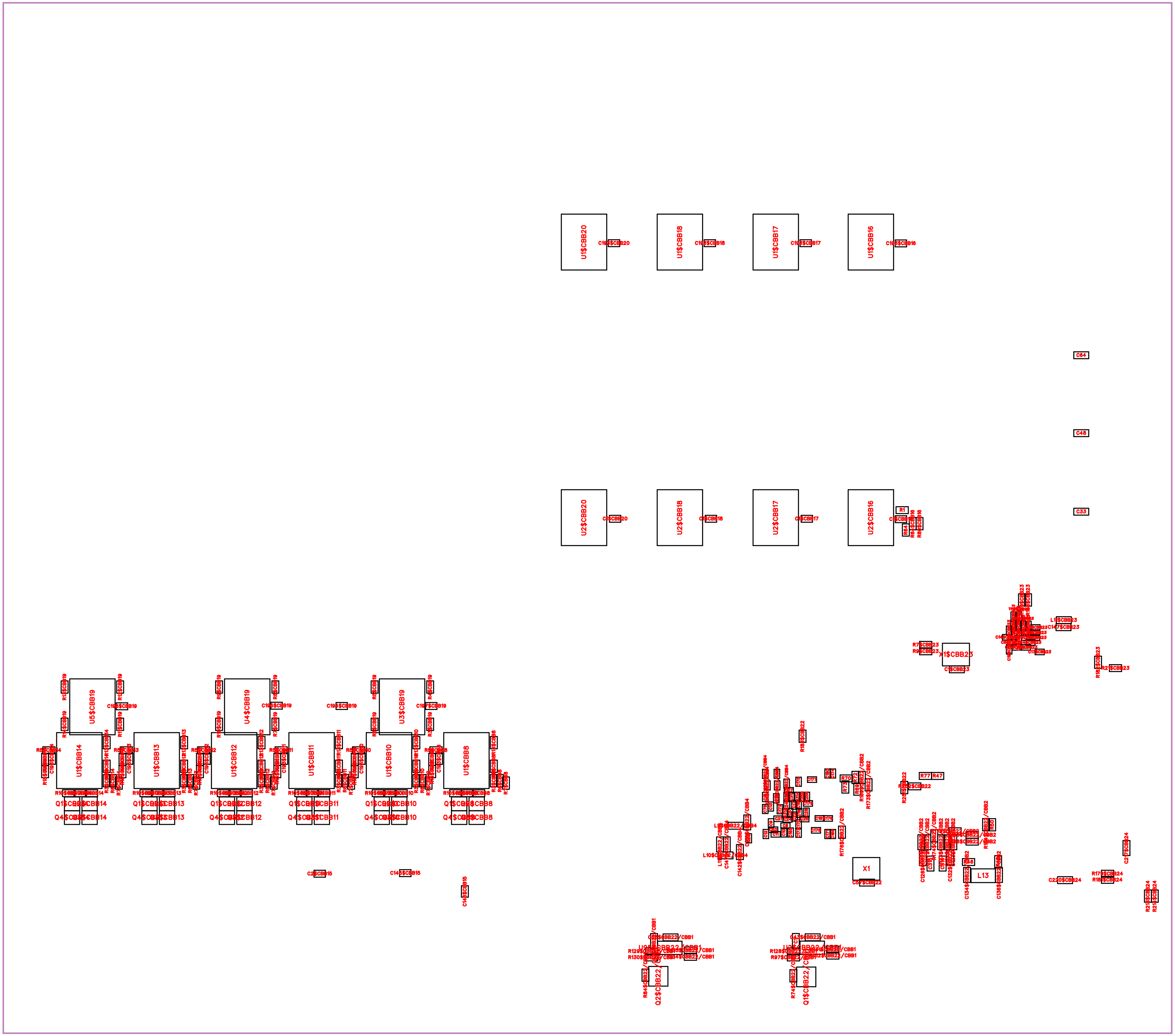
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No.	Quantity	Comment	Designator	Footprint	Value	Manufacturer Part	Manufacturer	Supplier Part	Supplier	Device	Name

1	328	100nF	C1$CBB7,C1$CBB9/CBB1,C1$CBB9/CBB2,C1$CBB9/CBB3,C1$CBB9/CBB4,C1$CBB9/CBB5,C1$CBB9/CBB6,C1$CBB9/CBB7,C1$CBB9/CBB8,C1$CBB9/CBB9,C1$CBB9/CBB10,C1$CBB9/CBB11,C1$CBB9/CBB12,C1$CBB9/CBB13,C1$CBB9/CBB14,C1$CBB9/CBB15,C1$CBB9/CBB16,C1$CBB9/CBB17,C1$CBB9/CBB18,C1$CBB9/CBB19,C1$CBB9/CBB20,C1$CBB9/CBB21,C1$CBB9/CBB22,C1$CBB9/CBB23,C1$CBB9/CBB24,C1$CBB9/CBB25,C1$CBB9/CBB26,C1$CBB9/CBB27,C1$CBB9/CBB28,C1$CBB9/CBB29,C1$CBB9/CBB30,C1$CBB9/CBB31,C1$CBB9/CBB32,C1$CBB9/CBB33,C1$CBB9/CBB34,C1$CBB9/CBB35,C1$CBB9/CBB36,C1$CBB9/CBB37,C1$CBB9/CBB38,C1$CBB9/CBB39,C1$CBB9/CBB40,C1$CBB9/CBB41,C1$CBB9/CBB42,C1$CBB16,C1$CBB17,C1$CBB18,C1$CBB19/CBB1,C1$CBB19/CBB2,C1$CBB19/CBB3,C1$CBB19/CBB4,C1$CBB19/CBB5,C1$CBB19/CBB6,C1$CBB19/CBB7,C1$CBB19/CBB8,C1$CBB19/CBB9,C1$CBB19/CBB10,C1$CBB19/CBB11,C1$CBB19/CBB12,C1$CBB19/CBB13,C1$CBB19/CBB14,C1$CBB19/CBB15,C1$CBB19/CBB16,C1$CBB19/CBB17,C1$CBB19/CBB18,C1$CBB19/CBB19,C1$CBB19/CBB20,C1$CBB19/CBB21,C1$CBB19/CBB22,C1$CBB19/CBB23,C1$CBB19/CBB24,C1$CBB19/CBB25,C1$CBB19/CBB26,C1$CBB19/CBB27,C1$CBB19/CBB28,C1$CBB19/CBB29,C1$CBB19/CBB30,C1$CBB19/CBB31,C1$CBB19/CBB32,C1$CBB19/CBB33,C1$CBB19/CBB34,C1$CBB19/CBB35,C1$CBB19/CBB36,C1$CBB20,C1$CBB21,C1$CBB49/CBB1,C1$CBB49/CBB2,C1$CBB49/CBB3,C1$CBB49/CBB4,C1$CBB49/CBB5,C1$CBB49/CBB6,C1$CBB49/CBB7,C1$CBB49/CBB8,C1$CBB49/CBB9,C1$CBB49/CBB10,C1$CBB49/CBB11,C1$CBB49/CBB12,C1$CBB49/CBB13,C1$CBB49/CBB14,C1$C	C0402	100nF	CL05B104KO5NNNC	SAMSUNG(三星)	C1525	LCSC	CL05B104KO5NNNC	100nF

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3	1	1uF	C2	C1206	1uF	CL31B105KBHNNNE	SAMSUNG(三星)	C1848	LCSC	CL31B105KBHNNNE	1uF

4	180	2.2nF	C2$CBB9/CBB1,C2$CBB9/CBB2,C2$CBB9/CBB3,C2$CBB9/CBB4,C2$CBB9/CBB5,C2$CBB9/CBB6,C2$CBB9/CBB7,C2$CBB9/CBB8,C2$CBB9/CBB9,C2$CBB9/CBB10,C2$CBB9/CBB11,C2$CBB9/CBB12,C2$CBB9/CBB13,C2$CBB9/CBB14,C2$CBB9/CBB15,C2$CBB9/CBB16,C2$CBB9/CBB17,C2$CBB9/CBB18,C2$CBB9/CBB19,C2$CBB9/CBB20,C2$CBB9/CBB21,C2$CBB9/CBB22,C2$CBB9/CBB23,C2$CBB9/CBB24,C2$CBB9/CBB25,C2$CBB9/CBB26,C2$CBB9/CBB27,C2$CBB9/CBB28,C2$CBB9/CBB29,C2$CBB9/CBB30,C2$CBB9/CBB31,C2$CBB9/CBB32,C2$CBB9/CBB33,C2$CBB9/CBB34,C2$CBB9/CBB35,C2$CBB9/CBB36,C2$CBB9/CBB37,C2$CBB9/CBB38,C2$CBB9/CBB39,C2$CBB9/CBB40,C2$CBB9/CBB41,C2$CBB9/CBB42,C2$CBB49/CBB1,C2$CBB49/CBB2,C2$CBB49/CBB3,C2$CBB49/CBB4,C2$CBB49/CBB5,C2$CBB49/CBB6,C2$CBB49/CBB7,C2$CBB49/CBB8,C2$CBB49/CBB9,C2$CBB49/CBB10,C2$CBB49/CBB11,C2$CBB49/CBB12,C2$CBB49/CBB13,C2$CBB49/CBB14,C2$CBB49/CBB15,C2$CBB49/CBB16,C2$CBB49/CBB17,C2$CBB49/CBB18,C2$CBB49/CBB19,C2$CBB49/CBB20,C2$CBB49/CBB21,C2$CBB49/CBB22,C2$CBB49/CBB23,C2$CBB49/CBB24,C2$CBB49/CBB25,C2$CBB49/CBB26,C2$CBB49/CBB27,C2$CBB49/CBB28,C2$CBB49/CBB29,C2$CBB49/CBB30,C2$CBB49/CBB31,C2$CBB49/CBB32,C2$CBB49/CBB33,C2$CBB49/CBB34,C2$CBB49/CBB35,C2$CBB49/CBB36,C2$CBB49/CBB37,C2$CBB49/CBB38,C2$CBB49/CBB39,C2$CBB49/CBB40,C2$CBB49/CBB41,C2$CBB49/CBB42,C2$CBB49/CBB43,C2$CBB49/CBB44,C2$CBB49/CBB45,C2$CBB49/CBB46,C2$CBB49/CBB47,C2$CBB49/CBB48,C4$CBB9/CBB1,C4$CBB9/CBB2,C4$CBB9/CBB3,C4$CBB9/CBB4,C4$CBB9/CBB5,C4$CBB9/CBB6	C0402	2.2nF	0402B222K500NT	FH(风华)	C1531	LCSC	0402B222K500NT	2.2nF	

5	78	100nF	C3$CBB23,C4$CBB23,C5,C5$CBB23,C6$CBB23,C7$CBB23,C8,C8$CBB23,C9,C9$CBB23,C10,C10$CBB23,C11,C11$CBB23,C12,C12$CBB23,C13,C13$CBB23,C14,C14$CBB23,C15,C15$CBB23,C16,C16$CBB23,C17,C17$CBB23,C18,C18$CBB23,C19,C19$CBB23,C20,C20$CBB23,C21,C21$CBB23,C22,C22$CBB22/CBB4,C22$CBB23,C23,C23$CBB22/CBB4,C23$CBB23,C24,C24$CBB22/CBB4,C25,C25$CBB22/CBB4,C26,C27,C28,C52,C78,C168,C169,C170,C171,C172,C173,C174,C175,C176,C177,C178,C179,C180,C181,C186$CBB24,C188$CBB24,C213$CBB24,C214$CBB24,C218$CBB24,C219$CBB24,C221$CBB24,C222$CBB24,C223$CBB24,C224$CBB24,C248,C250,C258,C313,C314	C0201	100nF	CL03A104KP3NNNC	SAMSUNG(三星)	C49062	LCSC	CL03A104KP3NNNC	100nF	
6	5	22uF	C3,C4,C7,C35,C72	C2220	22uF	FS55X226K500EHG	PSA(信昌电陶)	C784103	LCSC	FS55X226K500EHG	22uF	
7	12	100pF	C5$CBB7,C5$CBB21,C6$CBB7,C6$CBB21,C8$CBB7,C8$CBB21,C9$CBB7,C9$CBB21,C10$CBB7,C10$CBB21,C11$CBB7,C11$CBB21	C0402	100pF	0402CG101J500NT	FH(风华)	C1546	LCSC	0402CG101J500NT	100pF	
8	10	4.7uF	C6,C31,C38,C43,C60,C67,C212$CBB24,C215$CBB24,C217$CBB24,C220$CBB24	C0402	4.7uF	CL05A475MP5NRNC	SAMSUNG(三星)	C23733	LCSC	CL05A475MP5NRNC	4.7uF	
9	13	47uF	C29,C30,C32,C32$CBB23,C37,C39,C44,C50,C51,C53,C61,C66,C71	C1206	47uF	CL31A476MPHNNNE	SAMSUNG(三星)	C96123	LCSC	CL31A476MPHNNNE	47uF	
10	7	4.7nF	C33,C40,C45,C48,C54,C64,C68	C0402	4.7nF	0402B472K500NT	FH(风华)	C1538	LCSC	0402B472K500NT	4.7nF	
11	5	12pF	C34,C49,C55$CBB22/CBB2,C56$CBB22/CBB2,C65	C0402	12pF	V120J0402HQC500NBT	VIIYONG(微容)	C5251212	LCSC	V120J0402HQC500NBT	12pF	
12	3	1uF	C41,C57,C70	C0402	1uF	CL05A105KA5NQNC	SAMSUNG(三星)	C52923	LCSC	CL05A105KA5NQNC	1uF	
13	3	3.3nF	C56,C62,C73	C0402	3.3nF	CL05B332KB5NNNC	SAMSUNG(三星)	C26404	LCSC	CL05B332KB5NNNC	3.3nF	
14	3	47pF	C58,C69,C74	C0402	47pF	0402CG470J500NT	FH(风华)	C1567	LCSC	0402CG470J500NT	47pF	
15	5	330uF	C75,C76,C77,C79,C80	CAP-SMD_BD6.3-L6.6-W6.6-FD	330uF	RVT0J331M0607	HONOR(荣誉)	C58276	LCSC	RVT0J331M0607_C58276	330uF	

16	2	100uF	C81,C83	CAP-SMD_BD10.0-L10.3-W10.3-LS11.0-FD	100uF	MVK 50V100M 10*10	SAMYOUNG(韩国三莹)	C165627	LCSC	MVK 50V100M 10*10	100uF
17	1	330uF	C82	CAP-SMD_BD8.0-L8.3-W8.3-FD	330uF	RVT1E331M0810 330UF 25V	DMBJ(振宝佳)	C970701	LCSC	RVT1E331M0810 330UF 25V	330uF
18	4	4.7uF	C121$CBB22/CBB2,C122$CBB22/CBB2,C134$CBB22/CBB2,C136$CBB22/CBB2	C0402	4.7uF	LMK105BBJ475MVLF	TAIYO YUDEN(太诱)	C23733	LCSC	LMK105BBJ475MVLF	4.7uF
19	4	10uF	C138$CBB22/CBB4,C142$CBB22/CBB4,C147$CBB22/CBB4,C147$CBB23	C0402	10uF	CL05A106MQ5NUNC	SAMSUNG(三星)	C15525	LCSC	CL05A106MQ5NUNC	10uF
20	2	10pF	C184$CBB24,C185$CBB24	C0402	10pF	CL05C100JB5NNNC	SAMSUNG(三星)	C32949	LCSC	CL05C100JB5NNNC	10pF
21	1	USBLC6-2SC6	D1$CBB24	SOT-23-6_L2.9-W1.6-P0.95-LS2.8-BL		USBLC6-2SC6	ST(意法半导体)	C7519	LCSC	USBLC6-2SC6_C7519	USBLC6-2SC6
22	1	RBR5L60ADDTE25	D19	SMA_L4.5-W2.6-LS5.0-RD		RBR5L60ADDTE25	ROHM(罗姆)	C5343112	LCSC	RBR5L60ADDTE25	RBR5L60ADDTE25
23	32	KT-0603R	ETH-LED1,LED1,LED1$CBB15,LED1$CBB22,LED1$CBB23,LED2,LED2$CBB15,LED2$CBB23,LED3,LED3$CBB15,LED3$CBB23,LED4$CBB15,LED4$CBB23,LED5,LED5$CBB15,LED5$CBB23,LED6$CBB15,LED6$CBB23,LED7$CBB15,LED7$CBB23,LED8$CBB15,LED8$CBB23,LED9,LED9$CBB15,LED9$CBB23,LED10$CBB15,LED10$CBB23,LED11$CBB15,LED11$CBB23,LED12$CBB15,LED12$CBB23,LED13$CBB23	LED0603-RD		KT-0603R	KENTO	C2286	LCSC	KT-0603R	KT-0603R
24	2	522711579	FPC1$CBB22/CBB1,FPC3$CBB22/CBB1	FPC-SMD_522711579		522711579	MOLEX	C7545783	LCSC	522711579	522711579
25	6	PZ254-1-03-Z-8.5	H1$CBB8,H1$CBB10,H1$CBB11,H1$CBB12,H1$CBB13,H1$CBB14	HDR-TH_03P-P2.54-V-M		PZ254-1-03-Z-8.5	HCTL(华灿天禄)	C2894926	LCSC	PZ254-1-03-Z-8.5_C2894926	PZ254-1-03-Z-8.5
26	4	1.5uH	L1,L4,L5,L7	IND-SMD_L7.8-W7.0_GCD75	1.5uH	CD75-1R5M1.5UH	DMBJ(振宝佳)	C2826688	LCSC	CD75-1R5M1.5UH	1.5uH
27	3	6.8uH	L2,L3,L6	IND-SMD_L6.6-W6.6	6.8uH	SMMS0630-6R8M	SXN(顺翔诺)	C128690	LCSC	SMMS0630-6R8M	6.8uH
28	1	HR911130C	L8	RJ45-TH_HR911130C		HR911130C	HANRUN(汉仁)	C50933	LCSC	HR911130C	HR911130C
29	4	BLM15AX221SN1D	L9$CBB22/CBB4,L10$CBB22/CBB4,L11$CBB22/CBB4,L11$CBB23	L0402		BLM15AX221SN1D	muRata(村田)	C76885	LCSC	BLM15AX221SN1D	BLM15AX221SN1D

30	1	4.7uH	L12	IND-SMD_L7.8-W7.0_SUMIDA_CD73	4.7uH	FCD75-4R7M	cjiang(长江微电)	C41382395	LCSC	FCD75-4R7M	4.7uH	
31	1	APBD2012-152WTD12	L13	L0805		APBD2012-152WTD12	APV(爱普微)	C50325953	LCSC	APBD2012-152WTD12	APBD2012-152WTD12	
32	48	MMBT3904WT1G	Q1$CBB1,Q1$CBB2,Q1$CBB3,Q1$CBB4,Q1$CBB5,Q1$CBB6,Q1$CBB8,Q1$CBB10,Q1$CBB11,Q1$CBB12,Q1$CBB13,Q1$CBB14,Q2$CBB1,Q2$CBB2,Q2$CBB3,Q2$CBB4,Q2$CBB5,Q2$CBB6,Q2$CBB8,Q2$CBB10,Q2$CBB11,Q2$CBB12,Q2$CBB13,Q2$CBB14,Q3$CBB1,Q3$CBB2,Q3$CBB3,Q3$CBB4,Q3$CBB5,Q3$CBB6,Q3$CBB8,Q3$CBB10,Q3$CBB11,Q3$CBB12,Q3$CBB13,Q3$CBB14,Q4$CBB1,Q4$CBB2,Q4$CBB3,Q4$CBB4,Q4$CBB5,Q4$CBB6,Q4$CBB8,Q4$CBB10,Q4$CBB11,Q4$CBB12,Q4$CBB13,Q4$CBB14	SC-70-3_L2.1-W1.3-P1.30-LS2.1-BR		MMBT3904WT1G	onsemi(安森美)	C58011	LCSC	MMBT3904WT1G	MMBT3904WT1G	

33	516	DOX3134A	Q1$CBB15,Q2$CBB15,Q3$CBB15,Q4$CBB15,Q5$CBB9/CBB1,Q5$CBB9/CBB2,Q5$CBB9/CBB3,Q5$CBB9/CBB4,Q5$CBB9/CBB5,Q5$CBB9/CBB6,Q5$CBB9/CBB7,Q5$CBB9/CBB8,Q5$CBB9/CBB9,Q5$CBB9/CBB10,Q5$CBB9/CBB11,Q5$CBB9/CBB12,Q5$CBB9/CBB13,Q5$CBB9/CBB14,Q5$CBB9/CBB15,Q5$CBB9/CBB16,Q5$CBB9/CBB17,Q5$CBB9/CBB18,Q5$CBB9/CBB19,Q5$CBB9/CBB20,Q5$CBB9/CBB21,Q5$CBB9/CBB22,Q5$CBB9/CBB23,Q5$CBB9/CBB24,Q5$CBB9/CBB25,Q5$CBB9/CBB26,Q5$CBB9/CBB27,Q5$CBB9/CBB28,Q5$CBB9/CBB29,Q5$CBB9/CBB30,Q5$CBB9/CBB31,Q5$CBB9/CBB32,Q5$CBB9/CBB33,Q5$CBB9/CBB34,Q5$CBB9/CBB35,Q5$CBB9/CBB36,Q5$CBB9/CBB37,Q5$CBB9/CBB38,Q5$CBB9/CBB39,Q5$CBB9/CBB40,Q5$CBB9/CBB41,Q5$CBB9/CBB42,Q5$CBB15,Q5$CBB19/CBB1,Q5$CBB19/CBB2,Q5$CBB19/CBB3,Q5$CBB19/CBB4,Q5$CBB19/CBB5,Q5$CBB19/CBB6,Q5$CBB19/CBB7,Q5$CBB19/CBB8,Q5$CBB19/CBB9,Q5$CBB19/CBB10,Q5$CBB19/CBB11,Q5$CBB19/CBB12,Q5$CBB19/CBB13,Q5$CBB19/CBB14,Q5$CBB19/CBB15,Q5$CBB19/CBB16,Q5$CBB19/CBB17,Q5$CBB19/CBB18,Q5$CBB19/CBB19,Q5$CBB19/CBB20,Q5$CBB19/CBB21,Q5$CBB19/CBB22,Q5$CBB19/CBB23,Q5$CBB19/CBB24,Q5$CBB19/CBB25,Q5$CBB19/CBB26,Q5$CBB19/CBB27,Q5$CBB19/CBB28,Q5$CBB19/CBB29,Q5$CBB19/CBB30,Q5$CBB19/CBB31,Q5$CBB19/CBB32,Q5$CBB19/CBB33,Q5$CBB19/CBB34,Q5$CBB19/CBB35,Q5$CBB19/CBB36,Q5$CBB49/CBB1,Q5$CBB49/CBB2,Q5$CBB49/CBB3,Q5$CBB49/CBB4,Q5$CBB49/CBB5,Q5$CBB49/CBB6,Q5$CBB49/CBB7,Q5$CBB49/CBB8,Q5$CBB49/CBB9,Q5$CBB49/CBB10,Q5$CBB49/CBB11,Q5$CBB49/CBB12,Q5$CBB49/CBB13,Q5$CBB49/CBB14,Q5$CBB4	SOT-323-3_L2.1-W1.3-P1.30-LS2.3-BR		DOX3134A	DOINGTER(杜因特)	C54349756	LCSC	DOX3134A	DOX3134A	

34	2	BSS138	Q1$CBB22/CBB1,Q2$CBB22/CBB1	SOT-23-3_L2.9-W1.3-P1.90-LS2.4-BR-1		BSS138	KUU	C3015220	LCSC	BSS138	BSS138
35	4	DO2320AA	Q1$CBB23,Q2$CBB23,Q3$CBB23,Q4$CBB23	SOT-23-3_L2.9-W1.6-P1.90-LS2.8-BR		DO2320AA	DOINGTER(杜因特)	C47018574	LCSC	DO2320AA	DO2320AA
36	6	300Ω	R1$CBB1,R1$CBB2,R1$CBB3,R1$CBB4,R1$CBB5,R1$CBB6	R0402	300Ω	0402WGF3000TCE	UNI-ROYAL(厚声)	C25102	LCSC	0402WGF3000TCE	300Ω
37	60	100Ω	R1$CBB7,R1$CBB21,R2$CBB7,R2$CBB21,R3$CBB7,R3$CBB9,R3$CBB19,R3$CBB21,R3$CBB49,R4$CBB7,R4$CBB9,R4$CBB19,R4$CBB21,R4$CBB49,R5$CBB7,R5$CBB9,R5$CBB19,R5$CBB21,R5$CBB49,R6$CBB7,R6$CBB9,R6$CBB19,R6$CBB21,R6$CBB49,R7$CBB9,R7$CBB19,R7$CBB49,R8$CBB9,R8$CBB19,R8$CBB49,R9$CBB9,R9$CBB19,R9$CBB49,R10$CBB1,R10$CBB2,R10$CBB3,R10$CBB4,R10$CBB5,R10$CBB6,R10$CBB8,R10$CBB9,R10$CBB10,R10$CBB11,R10$CBB12,R10$CBB13,R10$CBB14,R10$CBB19,R10$CBB49,R11$CBB9,R11$CBB19,R11$CBB49,R12$CBB9,R12$CBB19,R12$CBB49,R13$CBB9,R13$CBB19,R13$CBB49,R14$CBB9,R14$CBB19,R14$CBB49	R0402	100Ω	0402WGF1000TCE	UNI-ROYAL(厚声)	C25076	LCSC	0402WGF1000TCE	100Ω

38	240	1MΩ	R1$CBB9/CBB1,R1$CBB9/CBB2,R1$CBB9/CBB3,R1$CBB9/CBB4,R1$CBB9/CBB5,R1$CBB9/CBB6,R1$CBB9/CBB7,R1$CBB9/CBB8,R1$CBB9/CBB9,R1$CBB9/CBB10,R1$CBB9/CBB11,R1$CBB9/CBB12,R1$CBB9/CBB13,R1$CBB9/CBB14,R1$CBB9/CBB15,R1$CBB9/CBB16,R1$CBB9/CBB17,R1$CBB9/CBB18,R1$CBB9/CBB19,R1$CBB9/CBB20,R1$CBB9/CBB21,R1$CBB9/CBB22,R1$CBB9/CBB23,R1$CBB9/CBB24,R1$CBB9/CBB25,R1$CBB9/CBB26,R1$CBB9/CBB27,R1$CBB9/CBB28,R1$CBB9/CBB29,R1$CBB9/CBB30,R1$CBB9/CBB31,R1$CBB9/CBB32,R1$CBB9/CBB33,R1$CBB9/CBB34,R1$CBB9/CBB35,R1$CBB9/CBB36,R1$CBB9/CBB37,R1$CBB9/CBB38,R1$CBB9/CBB39,R1$CBB9/CBB40,R1$CBB9/CBB41,R1$CBB9/CBB42,R1$CBB19/CBB1,R1$CBB19/CBB2,R1$CBB19/CBB3,R1$CBB19/CBB4,R1$CBB19/CBB5,R1$CBB19/CBB6,R1$CBB19/CBB7,R1$CBB19/CBB8,R1$CBB19/CBB9,R1$CBB19/CBB10,R1$CBB19/CBB11,R1$CBB19/CBB12,R1$CBB19/CBB13,R1$CBB19/CBB14,R1$CBB19/CBB15,R1$CBB19/CBB16,R1$CBB19/CBB17,R1$CBB19/CBB18,R1$CBB19/CBB19,R1$CBB19/CBB20,R1$CBB19/CBB21,R1$CBB19/CBB22,R1$CBB19/CBB23,R1$CBB19/CBB24,R1$CBB19/CBB25,R1$CBB19/CBB26,R1$CBB19/CBB27,R1$CBB19/CBB28,R1$CBB19/CBB29,R1$CBB19/CBB30,R1$CBB19/CBB31,R1$CBB19/CBB32,R1$CBB19/CBB33,R1$CBB19/CBB34,R1$CBB19/CBB35,R1$CBB19/CBB36,R1$CBB49/CBB1,R1$CBB49/CBB2,R1$CBB49/CBB3,R1$CBB49/CBB4,R1$CBB49/CBB5,R1$CBB49/CBB6,R1$CBB49/CBB7,R1$CBB49/CBB8,R1$CBB49/CBB9,R1$CBB49/CBB10,R1$CBB49/CBB11,R1$CBB49/CBB12,R1$CBB49/CBB13,R1$CBB49/CBB14,R1$CBB49/CBB15,R1$CBB49/CBB16,R1$CBB49/CBB17,R1$CBB49/C	R0402	1MΩ	0402WGF1004TCE	UNI-ROYAL(厚声)	C26083	LCSC	0402WGF1004TCE	1MΩ

39	91	1kΩ	R1$CBB15,R4$CBB1,R4$CBB2,R4$CBB3,R4$CBB4,R4$CBB5,R4$CBB6,R4$CBB8,R4$CBB10,R4$CBB11,R4$CBB12,R4$CBB13,R4$CBB14,R5$CBB1,R5$CBB2,R5$CBB3,R5$CBB4,R5$CBB5,R5$CBB6,R5$CBB8,R5$CBB10,R5$CBB11,R5$CBB12,R5$CBB13,R5$CBB14,R5$CBB15,R6$CBB1,R6$CBB2,R6$CBB3,R6$CBB4,R6$CBB5,R6$CBB6,R6$CBB8,R6$CBB10,R6$CBB11,R6$CBB12,R6$CBB13,R6$CBB14,R7$CBB1,R7$CBB2,R7$CBB3,R7$CBB4,R7$CBB5,R7$CBB6,R7$CBB8,R7$CBB10,R7$CBB11,R7$CBB12,R7$CBB13,R7$CBB14,R9$CBB1,R9$CBB2,R9$CBB3,R9$CBB4,R9$CBB5,R9$CBB6,R9$CBB8,R9$CBB10,R9$CBB11,R9$CBB12,R9$CBB13,R9$CBB14,R9$CBB15,R13$CBB8,R13$CBB10,R13$CBB11,R13$CBB12,R13$CBB13,R13$CBB14,R13$CBB15,R15$CBB8,R15$CBB10,R15$CBB11,R15$CBB12,R15$CBB13,R15$CBB14,R85$CBB15,R86$CBB15,R89$CBB15,R90$CBB15,R93$CBB15,R94$CBB15,R97$CBB15,R98$CBB15,R101$CBB15,R102$CBB15,R105$CBB15,R106$CBB15,R109$CBB15,R113$CBB15,R218$CBB24	R0402	1kΩ	0402WGF1001TCE	UNI-ROYAL(厚声)	C11702	LCSC	0402WGF1001TCE	1kΩ	

40	270	20kΩ	R1$CBB16/CBB1,R1$CBB16/CBB2,R1$CBB16/CBB3,R1$CBB16/CBB4,R1$CBB16/CBB5,R1$CBB16/CBB6,R1$CBB17/CBB1,R1$CBB17/CBB2,R1$CBB17/CBB3,R1$CBB17/CBB4,R1$CBB17/CBB5,R1$CBB17/CBB6,R1$CBB17/CBB7,R1$CBB17/CBB8,R1$CBB18/CBB1,R1$CBB18/CBB2,R1$CBB18/CBB3,R1$CBB18/CBB4,R1$CBB18/CBB5,R1$CBB18/CBB6,R1$CBB18/CBB7,R1$CBB18/CBB8,R1$CBB20/CBB1,R1$CBB20/CBB2,R1$CBB20/CBB3,R1$CBB20/CBB4,R1$CBB20/CBB5,R1$CBB20/CBB6,R1$CBB20/CBB7,R1$CBB20/CBB8,R3$CBB16/CBB1,R3$CBB16/CBB2,R3$CBB16/CBB3,R3$CBB16/CBB4,R3$CBB16/CBB5,R3$CBB16/CBB6,R3$CBB17/CBB1,R3$CBB17/CBB2,R3$CBB17/CBB3,R3$CBB17/CBB4,R3$CBB17/CBB5,R3$CBB17/CBB6,R3$CBB17/CBB7,R3$CBB17/CBB8,R3$CBB18/CBB1,R3$CBB18/CBB2,R3$CBB18/CBB3,R3$CBB18/CBB4,R3$CBB18/CBB5,R3$CBB18/CBB6,R3$CBB18/CBB7,R3$CBB18/CBB8,R3$CBB20/CBB1,R3$CBB20/CBB2,R3$CBB20/CBB3,R3$CBB20/CBB4,R3$CBB20/CBB5,R3$CBB20/CBB6,R3$CBB20/CBB7,R3$CBB20/CBB8,R4$CBB16/CBB1,R4$CBB16/CBB2,R4$CBB16/CBB3,R4$CBB16/CBB4,R4$CBB16/CBB5,R4$CBB16/CBB6,R4$CBB17/CBB1,R4$CBB17/CBB2,R4$CBB17/CBB3,R4$CBB17/CBB4,R4$CBB17/CBB5,R4$CBB17/CBB6,R4$CBB17/CBB7,R4$CBB17/CBB8,R4$CBB18/CBB1,R4$CBB18/CBB2,R4$CBB18/CBB3,R4$CBB18/CBB4,R4$CBB18/CBB5,R4$CBB18/CBB6,R4$CBB18/CBB7,R4$CBB18/CBB8,R4$CBB20/CBB1,R4$CBB20/CBB2,R4$CBB20/CBB3,R4$CBB20/CBB4,R4$CBB20/CBB5,R4$CBB20/CBB6,R4$CBB20/CBB7,R4$CBB20/CBB8,R5$CBB16/CBB1,R5$CBB16/CBB2,R5$CBB16/CBB3,R5$CBB16/CBB4,R5$CBB16/CBB5,R5$CBB16/CBB6,R5$CBB17/CBB1,R5$CBB17/	R0402	20kΩ	ARG02BTC2002	Viking(光颉)	C2984413	LCSC	ARG02BTC2002	20kΩ

41	34	470Ω	R1$CBB22,R10$CBB15,R11$CBB15,R12$CBB15,R12$CBB23,R13$CBB23,R14$CBB15,R14$CBB23,R15$CBB15,R15$CBB23,R16$CBB15,R16$CBB23,R17$CBB15,R17$CBB23,R18$CBB15,R18$CBB23,R19$CBB15,R19$CBB23,R20$CBB15,R20$CBB23,R21$CBB15,R22$CBB15,R31$CBB23,R32$CBB23,R33$CBB23,R34$CBB23,R49,R51,R54,R61,R95$CBB22,R166$CBB22/CBB2,R168$CBB22/CBB2,R260$CBB22	R0402	470Ω	0402WGF4700TCE	UNI-ROYAL(厚声)	C25117	LCSC	0402WGF4700TCE	470Ω
42	39	4.7kΩ	R1$CBB22/CBB2,R16$CBB17,R16$CBB18,R16$CBB20,R48$CBB17,R48$CBB18,R48$CBB20,R50,R74$CBB22/CBB1,R80$CBB17,R80$CBB18,R80$CBB20,R84$CBB22/CBB1,R97$CBB22/CBB1,R112$CBB17,R112$CBB18,R112$CBB20,R128$CBB22/CBB1,R129$CBB17,R129$CBB18,R129$CBB20,R129$CBB22/CBB1,R130,R130$CBB22/CBB1,R131$CBB17,R131$CBB18,R131$CBB20,R133$CBB17,R133$CBB18,R133$CBB20,R135$CBB17,R135$CBB20,R137,R165$CBB22/CBB2,R167$CBB22/CBB2,R169$CBB22/CBB2,R179$CBB22/CBB2,R179$CBB24,R180$CBB22/CBB2	R0402	4.7kΩ	0402WGF4701TCE	UNI-ROYAL(厚声)	C25900	LCSC	0402WGF4701TCE	4.7kΩ
43	28	10kΩ	R1$CBB23,R2$CBB23,R3$CBB23,R4$CBB23,R5$CBB23,R6$CBB23,R7$CBB23,R11$CBB23,R23$CBB23,R24$CBB23,R25$CBB23,R26$CBB23,R66,R67,R68,R69,R70,R71,R124,R139,R181$CBB24,R211$CBB24,R213$CBB24,R214$CBB24,R220$CBB24,R222$CBB24,R252$CBB22,R253$CBB22	R0402	10kΩ	0402WGF1002TCE	UNI-ROYAL(厚声)	C25744	LCSC	0402WGF1002TCE	10kΩ
44	12	330kΩ	R1$CBB49/CBB43,R1$CBB49/CBB44,R1$CBB49/CBB45,R1$CBB49/CBB46,R1$CBB49/CBB47,R1$CBB49/CBB48,R3$CBB49/CBB43,R3$CBB49/CBB44,R3$CBB49/CBB45,R3$CBB49/CBB46,R3$CBB49/CBB47,R3$CBB49/CBB48	R0402	330kΩ	RTT02334JTH	RALEC(旺诠)	C11887	LCSC	RTT02334JTH	330kΩ

45	4	20kΩ	R1,R64,R64$CBB16,R96$CBB16	R0402	20kΩ	0402WGF2002TCE	UNI-ROYAL(厚声)	C25765	LCSC	0402WGF2002TCE	20kΩ
46	12	1.2kΩ	R2$CBB1,R2$CBB2,R2$CBB3,R2$CBB4,R2$CBB5,R2$CBB6,R2$CBB8,R2$CBB10,R2$CBB11,R2$CBB12,R2$CBB13,R2$CBB14	R0402	1.2kΩ	0402WGF1201TCE	UNI-ROYAL(厚声)	C25862	LCSC	0402WGF1201TCE	1.2kΩ
47	6	3.9kΩ	R2$CBB15,R3$CBB15,R4$CBB15,R6$CBB15,R7$CBB15,R8$CBB15	R0402	3.9kΩ	0402WGF3901TCE	UNI-ROYAL(厚声)	C51721	LCSC	0402WGF3901TCE	3.9kΩ

48	210	10kΩ	R2$CBB16/CBB1,R2$CBB16/CBB2,R2$CBB16/CBB3,R2$CBB16/CBB4,R2$CBB16/CBB5,R2$CBB16/CBB6,R2$CBB17/CBB1,R2$CBB17/CBB2,R2$CBB17/CBB3,R2$CBB17/CBB4,R2$CBB17/CBB5,R2$CBB17/CBB6,R2$CBB17/CBB7,R2$CBB17/CBB8,R2$CBB18/CBB1,R2$CBB18/CBB2,R2$CBB18/CBB3,R2$CBB18/CBB4,R2$CBB18/CBB5,R2$CBB18/CBB6,R2$CBB18/CBB7,R2$CBB18/CBB8,R2$CBB20/CBB1,R2$CBB20/CBB2,R2$CBB20/CBB3,R2$CBB20/CBB4,R2$CBB20/CBB5,R2$CBB20/CBB6,R2$CBB20/CBB7,R2$CBB20/CBB8,R10$CBB16/CBB1,R10$CBB16/CBB2,R10$CBB16/CBB3,R10$CBB16/CBB4,R10$CBB16/CBB5,R10$CBB16/CBB6,R10$CBB17/CBB1,R10$CBB17/CBB2,R10$CBB17/CBB3,R10$CBB17/CBB4,R10$CBB17/CBB5,R10$CBB17/CBB6,R10$CBB17/CBB7,R10$CBB17/CBB8,R10$CBB18/CBB1,R10$CBB18/CBB2,R10$CBB18/CBB3,R10$CBB18/CBB4,R10$CBB18/CBB5,R10$CBB18/CBB6,R10$CBB18/CBB7,R10$CBB18/CBB8,R10$CBB20/CBB1,R10$CBB20/CBB2,R10$CBB20/CBB3,R10$CBB20/CBB4,R10$CBB20/CBB5,R10$CBB20/CBB6,R10$CBB20/CBB7,R10$CBB20/CBB8,R11$CBB16/CBB1,R11$CBB16/CBB2,R11$CBB16/CBB3,R11$CBB16/CBB4,R11$CBB16/CBB5,R11$CBB16/CBB6,R11$CBB17/CBB1,R11$CBB17/CBB2,R11$CBB17/CBB3,R11$CBB17/CBB4,R11$CBB17/CBB5,R11$CBB17/CBB6,R11$CBB17/CBB7,R11$CBB17/CBB8,R11$CBB18/CBB1,R11$CBB18/CBB2,R11$CBB18/CBB3,R11$CBB18/CBB4,R11$CBB18/CBB5,R11$CBB18/CBB6,R11$CBB18/CBB7,R11$CBB18/CBB8,R11$CBB20/CBB1,R11$CBB20/CBB2,R11$CBB20/CBB3,R11$CBB20/CBB4,R11$CBB20/CBB5,R11$CBB20/CBB6,R11$CBB20/CBB7,R11$CBB20/CBB8,R12$CBB16/CBB1,R12$CBB16/CBB2,R12$CBB16/CBB3,R12$	R0402	10kΩ	RT0402BRD0710KL	YAGEO(国巨)	C190095	LCSC	RT0402BRD0710KL	10kΩ

49	60	120Ω	R2,R4,R5,R7,R8,R10,R11,R13,R14,R16,R17,R19,R20,R22,R23,R25,R26,R28,R29,R31,R32,R34,R35,R37,R38,R40,R41,R43,R44,R63,R74,R80,R81,R83,R84,R86,R87,R89,R90,R92,R93,R95,R97,R99,R100,R102,R103,R105,R106,R108,R109,R111,R112,R114,R115,R117,R118,R120,R121,R123	R0402	120Ω	0402WGF1200TCE	UNI-ROYAL(厚声)	C25079	LCSC	0402WGF1200TCE	120Ω
50	24	910Ω	R3$CBB1,R3$CBB2,R3$CBB3,R3$CBB4,R3$CBB5,R3$CBB6,R3$CBB8,R3$CBB10,R3$CBB11,R3$CBB12,R3$CBB13,R3$CBB14,R8$CBB1,R8$CBB2,R8$CBB3,R8$CBB4,R8$CBB5,R8$CBB6,R8$CBB8,R8$CBB10,R8$CBB11,R8$CBB12,R8$CBB13,R8$CBB14	R0402	910Ω	SCR0402J910R	VO(翔胜)	C3017583	LCSC	SCR0402J910R	910Ω
51	30	68Ω	R3,R6,R9,R12,R15,R18,R21,R24,R27,R30,R33,R36,R39,R42,R62,R79,R82,R85,R88,R91,R94,R98,R101,R104,R107,R110,R113,R116,R119,R122	R0402	68Ω	0402WGJ0680TCE	UNI-ROYAL(厚声)	C25178	LCSC	0402WGJ0680TCE	68Ω
52	30	3.3kΩ	R11$CBB1,R11$CBB2,R11$CBB3,R11$CBB4,R11$CBB5,R11$CBB6,R12$CBB1,R12$CBB2,R12$CBB3,R12$CBB4,R12$CBB5,R12$CBB6,R13$CBB1,R13$CBB2,R13$CBB3,R13$CBB4,R13$CBB5,R13$CBB6,R87$CBB15,R88$CBB15,R91$CBB15,R92$CBB15,R95$CBB15,R96$CBB15,R99$CBB15,R100$CBB15,R103$CBB15,R104$CBB15,R107$CBB15,R108$CBB15	R0402	3.3kΩ	0402WGF3301TCE	UNI-ROYAL(厚声)	C25890	LCSC	0402WGF3301TCE	3.3kΩ
53	6	270Ω	R14$CBB1,R14$CBB2,R14$CBB3,R14$CBB4,R14$CBB5,R14$CBB6	R0402	270Ω	CRCW0402270RFKED	VISHAY(威世)	C482120	LCSC	CRCW0402270RFKED	270Ω
54	7	22Ω	R21$CBB23,R22$CBB23,R77,R96,R135,R149,R182$CBB23	R0402	22Ω	0402WGF220JTCE	UNI-ROYAL(厚声)	C25092	LCSC	0402WGF220JTCE	22Ω
55	4	33Ω	R27$CBB23,R28$CBB23,R29$CBB23,R30$CBB23	R2512	33Ω	CRH2512F33R0E04Z	EVER OHMS(天二科技)	C175260	LCSC	CRH2512F33R0E04Z	33Ω
56	8	1.5kΩ	R32$CBB16,R98$CBB16,R100$CBB16,R102$CBB16,R104$CBB16,R106$CBB16,R134,R142	R0402	1.5kΩ	0402WGF1501TCE	UNI-ROYAL(厚声)	C25867	LCSC	0402WGF1501TCE	1.5kΩ

57	23	51kΩ	R32$CBB17,R32$CBB18,R32$CBB20,R64$CBB17,R64$CBB18,R64$CBB20,R96$CBB17,R96$CBB18,R96$CBB20,R128$CBB17,R128$CBB18,R128$CBB20,R130$CBB17,R130$CBB18,R130$CBB20,R132$CBB17,R132$CBB18,R132$CBB20,R134$CBB17,R134$CBB18,R134$CBB20,R136$CBB17,R136$CBB20	R0402	51kΩ	0402WGF5102TCE	UNI-ROYAL(厚声)	C25794	LCSC	0402WGF5102TCE	51kΩ
58	8	0Ω	R45,R65,R148,R158$CBB22,R158$CBB23,R159$CBB23,R164$CBB22/CBB2,R174$CBB22/CBB2	R0402	0Ω	0402WGF0000TCE	UNI-ROYAL(厚声)	C17168	LCSC	0402WGF0000TCE	0Ω
59	1	6.49kΩ	R46	R0402	6.49kΩ	RT0402BRE076K49L	YAGEO(国巨)	C853380	LCSC	RT0402BRE076K49L	6.49kΩ
60	7	1.8kΩ	R48$CBB16,R99$CBB16,R101$CBB16,R103$CBB16,R105$CBB16,R107$CBB16,R212$CBB24	R0402	1.8kΩ	0402WGF1801TCE	UNI-ROYAL(厚声)	C25871	LCSC	0402WGF1801TCE	1.8kΩ
61	3	12kΩ	R51$CBB23,R78,R183$CBB22	R0402	12kΩ	0402WGF1202TCE	UNI-ROYAL(厚声)	C25752	LCSC	0402WGF1202TCE	12kΩ
62	3	15kΩ	R52,R136,R138	R0402	15kΩ	0402WGF1502TCE	UNI-ROYAL(厚声)	C25756	LCSC	0402WGF1502TCE	15kΩ
63	3	560kΩ	R53,R128,R140	R0402	560kΩ	0402WGF5603TCE	UNI-ROYAL(厚声)	C132339	LCSC	0402WGF5603TCE	560kΩ
64	1	8.2kΩ	R55	R0402	8.2kΩ	CR0402FF8201G	LIZ(丽智电子)	C100538	LCSC	CR0402FF8201G	8.2kΩ
65	3	40.2kΩ	R56,R129,R143	R0402	40.2kΩ	0402WGF4022TCE	UNI-ROYAL(厚声)	C25893	LCSC	0402WGF4022TCE	40.2kΩ
66	1	24kΩ	R57	R0402	24kΩ	0402WGF2402TCE	UNI-ROYAL(厚声)	C25769	LCSC	0402WGF2402TCE	24kΩ
67	4	0Ω	R58,R58$CBB23,R132,R146	R1210	0Ω	RS-1210000JT	FH(风华)	C140000	LCSC	RS-1210000JT	0Ω
68	1	19.1kΩ	R59	R0402	19.1kΩ	0402WGF1912TCE	UNI-ROYAL(厚声)	C52036	LCSC	0402WGF1912TCE	19.1kΩ
69	2	36kΩ	R60,R131	R0402	36kΩ	0402WGJ0363TCE	UNI-ROYAL(厚声)	C25556	LCSC	0402WGJ0363TCE	36kΩ
70	2	27kΩ	R80$CBB16,R97$CBB16	R0402	27kΩ	0402WGF2702TCE	UNI-ROYAL(厚声)	C25771	LCSC	0402WGF2702TCE	27kΩ
71	1	43kΩ	R125	R0402	43kΩ	0402WGJ0433TCE	UNI-ROYAL(厚声)	C25561	LCSC	0402WGJ0433TCE	43kΩ
72	2	82kΩ	R126,R147	R0402	82kΩ	0402WGF8202TCE	UNI-ROYAL(厚声)	C4142	LCSC	0402WGF8202TCE	82kΩ
73	2	12kΩ	R127,R219$CBB24	R0402	12kΩ	0402WGJ0123TCE	UNI-ROYAL(厚声)	C25535	LCSC	0402WGJ0123TCE	12kΩ
74	1	32.4kΩ	R133	R0402	32.4kΩ	0402WGF3242TCE	UNI-ROYAL(厚声)	C26974	LCSC	0402WGF3242TCE	32.4kΩ

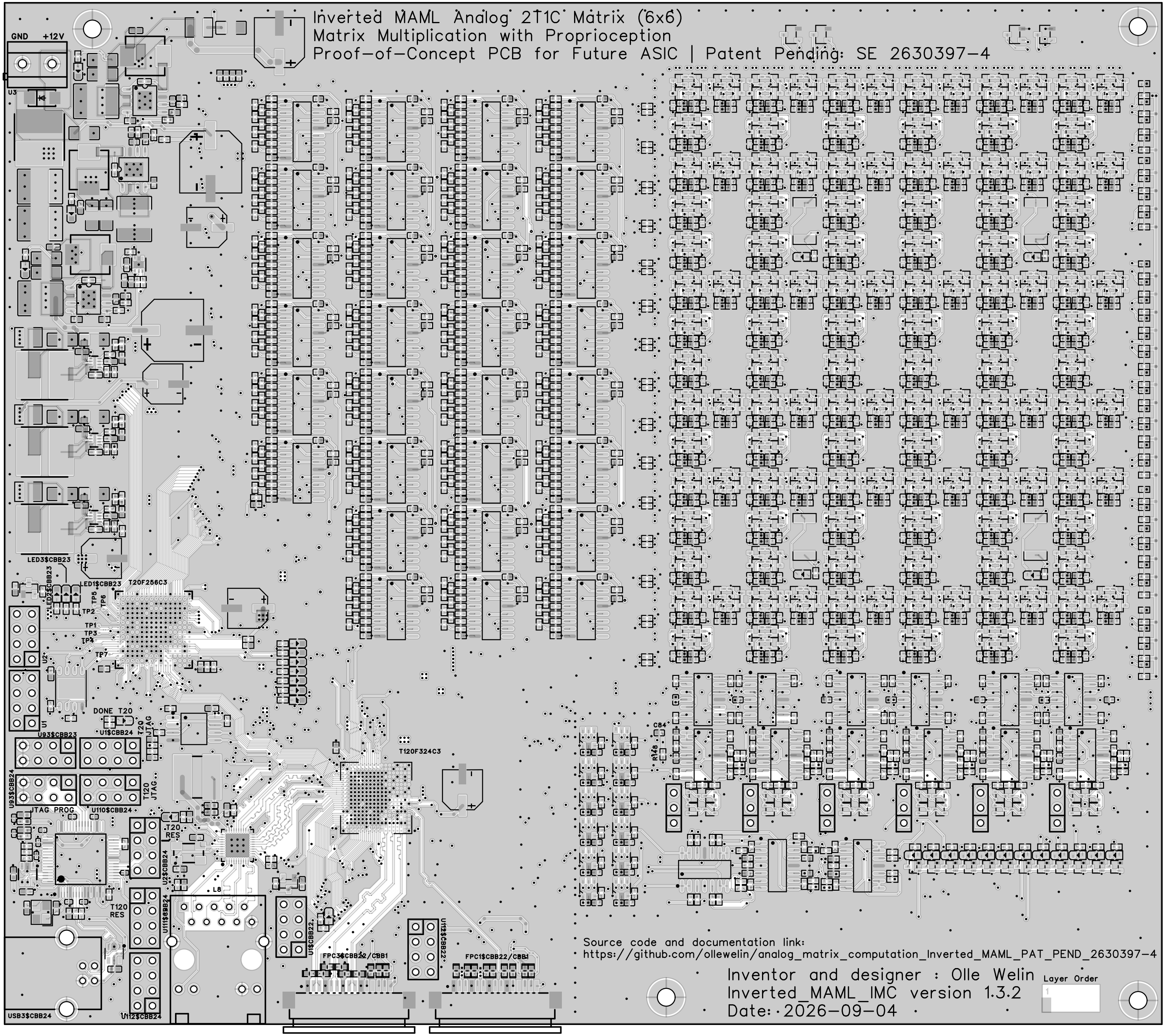
75	4	68kΩ	R141,R144,R150,R151	R0402	68kΩ	0402WGF6802TCE	UNI-ROYAL(厚声)	C36871	LCSC	0402WGF6802TCE	68kΩ
76	1	75kΩ	R145	R0402	75kΩ	0402WGF7502TCE	UNI-ROYAL(厚声)	C25798	LCSC	0402WGF7502TCE	75kΩ
77	2	10Ω	R215$CBB24,R217$CBB24	R0402	10Ω	0402WGF100JTCE	UNI-ROYAL(厚声)	C25077	LCSC	0402WGF100JTCE	10Ω
78	9	Test-Point	TP1,TP2,TP2$CBB23,TP3,TP3$CBB23,TP4,TP5,TP6,TP7	Test-Point-0.5mm						Test-Point	
79	32	MCP6024T-I/SL-MS	U1$CBB1,U1$CBB2,U1$CBB3,U1$CBB4,U1$CBB5,U1$CBB6,U1$CBB8,U1$CBB10,U1$CBB11,U1$CBB12,U1$CBB13,U1$CBB14,U1$CBB15,U1$CBB16,U1$CBB17,U1$CBB18,U1$CBB20,U2$CBB15,U2$CBB16,U2$CBB17,U2$CBB18,U2$CBB20,U3$CBB9,U3$CBB15,U3$CBB19,U3$CBB49,U4$CBB9,U4$CBB19,U4$CBB49,U5$CBB9,U5$CBB19,U5$CBB49	SOIC-14_L8.7-W3.9-P1.27-LS6.0-BL-1		MCP6024T-I/SL-MS	MSKSEMI(美森科)	C54539958	LCSC	MCP6024T-I/SL-MS	MCP6024T-I/SL-MS
80	12	XC2361A	U1$CBB7,U1$CBB21,U2$CBB7,U2$CBB21,U3$CBB7,U3$CBB21,U4$CBB7,U4$CBB21,U5$CBB7,U5$CBB21,U7$CBB7,U7$CBB21	TSOT-23-6_L2.9-W1.6-P0.95-LS2.6-BL		XC2361A	SiChange(芯昌科技)	C41383383	LCSC	XC2361A	XC2361A
81	1	T20F256C3	U1$CBB23	FBGA-256_L13.0-W13.0-R16-C16-P0.80-TL		T20F256C3	Efinix	C3277960	LCSC	T20F256C3	T20F256C3
82	11	PZ2.54-2*4	U1,U1$CBB22,U1$CBB24,U2,U2$CBB24,U93$CBB23,U93$CBB24,U110$CBB24,U111$CBB24,U112$CBB22,U112$CBB24	HDR-TH_8P-P2.54-V-M-R2-C4-S2.54		PZ2.54-2*4	ZHOURI(洲日)	C5156676	LCSC	PZ2.54-2*4	PZ2.54-2*4
83	2	PCA9306DC1	U2$CBB22/CBB1,U95$CBB22/CBB1	VSSOP-8_L2.4-W2.1-P0.50-LS3.2-BR		PCA9306DC1	NXP(恩智浦)	C129510	LCSC	PCA9306DC1	PCA9306DC1
84	1	KF301-5.0-2P	U3	CONN-TH_P5.00_KF301-5.0-2P		KF301-5.0-2P	KEFA(科发)	C474881	LCSC	KF301-5.0-2P	KF301-5.0-2P
85	3	AP64350SP-13	U5,U6,U7	SO-8_L4.9-W3.9-P1.27-LS6.0-BL-EP		AP64350SP-13	DIODES(美台)	C2071691	LCSC	AP64350SP-13	AP64350SP-13

86	30	74HC595	U6$CBB16/CBB1,U6$CBB16/CBB2,U6$CBB16/CBB3,U6$CBB16/CBB4,U6$CBB16/CBB5,U6$CBB16/CBB6,U6$CBB17/CBB1,U6$CBB17/CBB2,U6$CBB17/CBB3,U6$CBB17/CBB4,U6$CBB17/CBB5,U6$CBB17/CBB6,U6$CBB17/CBB7,U6$CBB17/CBB8,U6$CBB18/CBB1,U6$CBB18/CBB2,U6$CBB18/CBB3,U6$CBB18/CBB4,U6$CBB18/CBB5,U6$CBB18/CBB6,U6$CBB18/CBB7,U6$CBB18/CBB8,U6$CBB20/CBB1,U6$CBB20/CBB2,U6$CBB20/CBB3,U6$CBB20/CBB4,U6$CBB20/CBB5,U6$CBB20/CBB6,U6$CBB20/CBB7,U6$CBB20/CBB8	SOP-16_L9.9-W3.9-P1.27-LS6.0-BL		74HC595	GR(国睿)	C18164493	LCSC	74HC595	74HC595
87	3	MP2393GTL-Z	U9,U12,U16	SOT-583-8_L2.1-W1.2-P0.50-LS1.6-BL		MP2393GTL-Z	MPS(芯源)	C6674609	LCSC	MP2393GTL-Z	MP2393GTL-Z
88	2	APX803S-31SA-7	U14,U118$CBB23	SOT-23-3_L2.9-W1.3-P1.90-LS2.4-BR		APX803S-31SA-7	DIODES(美台)	C129757	LCSC	APX803S-31SA-7	APX803S-31SA-7
89	1	W25Q64JVZPIQ	U19$CBB23	WSON-8_L6.0-W5.0-P1.27-BL-EP		W25Q64JVZPIQ	WINBOND(华邦)	C2940197	LCSC	W25Q64JVZPIQ	W25Q64JVZPIQ
90	1	RTL8211F-CG	U98$CBB22/CBB2	WQFN-40_L5.0-W5.0-P0.4-BL-EP		RTL8211F-CG	REALTEK(瑞昱)	C187932	LCSC	RTL8211F-CG	RTL8211F-CG
91	1	FT2232HL-REEL	U98$CBB24	LQFP-64_L10.0-W10.0-P0.50-LS12.0-TL		FT2232HL-REEL	FTDI(飞特帝亚)	C27882	LCSC	FT2232HL-REEL	FT2232HL-REEL
92	1	W25Q128JVSIQ	U99$CBB22	SOIC-8_L5.3-W5.3-P1.27-LS8.0-BL		W25Q128JVSIQ	Winbond(华邦)	C97521	LCSC	W25Q128JVSIQ	W25Q128JVSIQ
93	1	T120F324C3	U100$CBB22	BGA324C65P18X18_1200X1200X97N						T120F324C3	T120F324C3
94	1	12MHz	U101$CBB24	CRYSTAL-SMD_4P-L3.2-W2.5-BL	12MHz	ECS-120-10-33B-CKM-TR	ECS	C2450718	LCSC	ECS-120-10-33B-CKM-TR	12MHz
95	2	BLM15PX330SN1D	U103$CBB24,U104$CBB24	L0402		BLM15PX330SN1D	muRata(村田)	C160974	LCSC	BLM15PX330SN1D	BLM15PX330SN1D
96	1	USB-B10-BRW	USB3$CBB24	USB-B-TH_USB-B10-BRW		USB-B10-BRW	XUNPU(讯普)	C720549	LCSC	USB-B10-BRW	USB-B10-BRW
97	2	50MHz	X1,X1$CBB23	OSC-SMD_4P-L3.2-W2.5-BL	50MHz	OT322550MJBA4SL	YXC(扬兴晶振)	C717684	LCSC	OT322550MJBA4SL	50MHz

98	1	25MHz	X4$CBB22/CBB2	CRYSTAL-SMD_4P-L5.0-W3.2-BL	25MHz	XC53M4-25.000-F12JJDT	HCI(杭晶)	C2916916	LCSC	XC53M4-25.000-F12JJDT	25MHz

Current Rating	Current - Saturation(Isat)	Dielectric	Resistance	Voltage Rated	Voltage Rating
					16V
					50V
					10V
					50V
					50V
					10V
					10V
					50V
					50V
					25V
					50V
					50V
					6.3V
					50V
					25V
					6.3V
					50V
3A					
5.3A					
-	6A				
600mA					
	4.2A				
550mA					

3A					
					30V


Inverted MAML Analog 2T1C Matrix (6x6)
Matrix Multiplication with Proprioception
Proof-of-Concept PCB for Future ASIC | Patent Pending: SE 2630397-4
GND
+12V
U3
LED3$CBB23
LED1$CBB23
T20F256C3
TP2
TP1
TP3
TP7
DONE T20
U1$CBB24
U93$CBB23
T20 JTAG
T120 JTAG
JTAG PROG
U110$CBB24
T120F324C3
T20 RES
T120 RES
L8
C84
USB3$CBB24
U112$CBB24
FPC3$CBB22/CBB1
FPC1$CBB22/CBB1
Source code and documentation link:
https://github.com/ollewelin/analog_matrix_computation_Inverted_MAML_PAT_PEND_2630397-4
Inventor and designer : Olle Welin
Inverted_MAML_IMC version 1.3.2
Date: 2026-09-04
Layer Order